

Multi-Language Sentiment Analysis Using BERT and BiLSTM

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Abstract—The growing globalization of online communication requires the development of powerful multilingual sentiment analysis (MLSA) systems, which are able to decode the emotional settings of various languages and different cultures. In the presented paper, a hybrid architecture is proposed that involves Bidirectional Encoder Representations with Transformers (mBERT) and Bidirectional Long Short-Term Memory (BiLSTM) networks to improve multilingual sentiment classification. Instead of relying on contextual embeddings, or recurrent sequence modeling, the proposed BERT-BiLSTM fusion architecture uses the semantic richness of mBERT but pathways sequential dependencies with BiLSTM layers, which leads to cross-lingual emotion representation by using the weighted cross-entropy regularization and attention-based token alignment. Four benchmark multilingual datasets (Twitter-MLSA, MultiEmo, Amazon-Reviews-Mix, and Code-Mixed-SocialMedia) with ten languages, such as English, Hindi, Arabic, and Spanish are experimented on. Findings indicate a mean 6.4 per cent enhancement in F1-scores compared to state-of-the-art hybrid assistants like CNN-BiLSTM and XLM-RoBERTa-GRU.

Index Terms—Multilingual sentiment analysis, BERT, Bidirectional Long Short-Term Memory (BiLSTM), natural language processing (NLP), transformer models.

I. INTRODUCTION

Over the past years, the growth of user-created content in social media, review sites, and messaging apps has created enormous amounts of textual information across various languages in numerous languages, a lot of it unedited and unfiltered [1]. Knowledge of the feeling conveyed in such data as positive, negative or neutral is of significant interest to the marketing analytics, social monitoring as well as the place of mining public opinion [2]. Conventional research in sentiment analysis has mostly been done in the context of monolanguages and has frequently used lexical or classical machine-learning strategies. However, as digital communication becomes increasingly global and multilingual, there is a growing need for robust sentiment analysis systems that can operate across languages and cultural boundaries [3].

Multilingual sentiment analysis (MLSA) presents unique challenges beyond those of monolingual tasks. These include variation in syntax and vocabulary across languages, the scarcity of large annotated corpora for low-resource languages, and the nuances of cross-cultural expression of sentiment [4]. Moreover, code-mixed text where more than one language is used in the same sentence further complicates modeling. To

address these issues, recent work has turned to deep learning, and specifically to pretrained multilingual transformer models (such as multilingual BERT) combined with sequence-modeling architectures [5].

Among deep-learning approaches, the integration of a transformer model for contextualised embeddings and a recurrent neural network for sequential dependency modelling has shown promise in sentiment classification tasks. For example, Li et al. [6] introduced a BERT + BiLSTM architecture for Chinese buzzwords sentiment analysis and reported improved F1 scores compared to baseline models. Their work underlines the benefit of combining the deep contextualisation of BERT with the temporal modelling of BiLSTM. However, that study remains monolingual and does not address the multilingual or cross-lingual scenario.

This paper proposes a novel hybrid architecture that integrates a multilingual transformer (mBERT) with a Bidirectional Long Short-Term Memory (BiLSTM) network, specifically designed for MLSA across multiple languages including high-resource and low-resource languages, as well as code-mixed texts. Unlike prior work, our design incorporates a cross-lingual alignment layer that mitigates embedding drift between languages, and a language-adaptive regularisation loss to handle class imbalance and linguistic diversity. We demonstrate through extensive experiments that our approach outperforms existing multilingual baselines and provides improved generalisation in low-resource and code-mixed settings.

II. RELATED WORK AND LITERATURE REVIEW

Sentiment analysis (SA) has grown substantially in both monolingual and multilingual contexts. Early work focused on English and other high-resource languages, leveraging lexicon-based and classical machine-learning approaches [7]. For instance, Banea et al. [3] investigated multilingual subjectivity and sentiment in five European languages, identifying that differences in syntax and vocabulary impose substantial barriers across languages.

A. Multilingual Sentiment Analysis

Multilingual sentiment analysis (MLSA) introduces unique challenges: linguistic variation, cultural nuances, scarcity of annotated data for low-resource languages, and domain shifts

across languages [8]. Lo et al. [4] survey MLSA from formal to informal texts, highlighting the under-explored status of code-mixed and low-resource languages. More recently, Abdullah Rusli [1] provide a systematic literature review of MLSA, noting that while transformer models are increasingly used, transfer-learning and cross-lingual adaptation remain immature.

B. Deep-Learning Architectures for Sentiment Analysis

In the monolingual domain, deep learning has outperformed earlier methods by capturing rich contextual features. For example, Li, Lei Ji [6] apply a hybrid architecture combining BERT and BiLSTM for Chinese “buzzword” sentiment analysis, demonstrating notable F1-score gains. Their model fine-tunes BERT to yield dynamic embeddings, which are then passed through a BiLSTM to capture sequential polarity patterns [6].

In multilingual and cross-lingual settings, hybrid and ensemble models have begun to surface. Jain et al. [5] propose an “optimized hybrid deep-learning” model for cross-linguistic sentiment analysis, combining CNNs, LSTMs, and nature-inspired optimisation (Grey Wolf Optimisation) to handle multilingual data from diverse languages. While innovative, their approach does not specifically integrate large multilingual transformer encoders nor address code-mixed contexts explicitly.

C. Transformer-based and Hybrid Approaches

Transformer-based architectures, especially multilingual variants such as mBERT and XLM-R, have driven recent progress in MLSA. For example, a recent study by Khan et al. [9] targets Urdu sentiment analysis with a stacked CNN-BiLSTM network built on mBERT embeddings, indicating that rich pretrained multilingual embeddings help even in resource-poor languages. Similarly, shared-task work such as the SemEval-2023 Task 12 highlighted the value of pretrained multilingual models (Afro-XLMR, mBERT) for low-resource African languages [8], [10].

Despite these advances, a gap remains: Many studies apply transformer or recurrent models in isolation, or use them sequentially, but few integrate a transformer encoder + recurrent model + explicit cross-lingual alignment mechanism in a unified architecture for multilingual sentiment classification. Moreover, code-mixed and low-resource languages often remain under-explored.

MLSA must address significant linguistic variability and limited resources [11]. Hybrid models (transformer + recurrent) show strong promise in monolingual tasks but less so in fully multilingual or code-mixed scenarios [12]. While multilingual transformer models reduce language-dependency, adaptation to cultural nuance and sequential dependency (especially for sentiment) remains incomplete. Few works integrate a dedicated cross-lingual alignment or attention mechanism specifically to reduce embedding drift across languages [13].

These observations motivate the proposed architecture, which combines a multilingual transformer encoder, BiLSTM

layers, and a cross-lingual alignment module to support robust sentiment classification across high- and low-resource languages including code-mixed texts.

The key contributions of this work are:

- A hybrid transformer–recurrent architecture (mBERT + BiLSTM) tailored for multilingual sentiment classification.
- A novel cross-lingual embedding alignment mechanism to support transfer between languages with divergent syntax and morphology.
- Experimental validation on multiple benchmark multilingual and code-mixed datasets, showing significant performance gains over state-of-the-art baselines.

III. PROPOSED METHODOLOGY

This section presents the proposed hybrid architecture that integrates a multilingual transformer encoder (mBERT) with a Bidirectional Long Short-Term Memory (BiLSTM) network for robust multilingual sentiment analysis (MLSA). The model aims to learn cross-lingual semantic representations while maintaining sequential dependencies within text sequences. The overall workflow is illustrated in Fig. 1.

The proposed framework comprises four major components:

- 1) **Multilingual Embedding Layer:** Contextual token embeddings are extracted using multilingual BERT (mBERT), pretrained on 104 languages.
- 2) **BiLSTM Encoding Layer:** Sequential features are captured through bidirectional recurrence to preserve both past and future contextual information.
- 3) **Cross-Lingual Alignment Module:** An attention-based mechanism aligns language-specific embeddings into a shared latent space to mitigate feature drift across languages.
- 4) **Classification Layer:** A dense layer followed by a softmax activation predicts sentiment polarity across three or more classes (positive, negative, neutral).

Let $S = [w_1, w_2, \dots, w_T]$ denote an input sentence of T tokens in any language L_i . Each token w_t is embedded using mBERT into a contextualized vector $h_t \in \mathbb{R}^d$, where d is the embedding dimension:

$$h_t = \text{BERT}(w_t \mid \theta_{\text{BERT}}) \quad (1)$$

The sequence of embeddings $H = [h_1, h_2, \dots, h_T]$ is then passed to a bidirectional LSTM to model sequential dependencies in both directions:

$$\vec{h}_t = \text{LSTM}_{fwd}(h_t, \vec{h}_{t-1}) \quad (2)$$

$$\overleftarrow{h}_t = \text{LSTM}_{bwd}(h_t, \overleftarrow{h}_{t+1}) \quad (3)$$

$$H_t = [\vec{h}_t; \overleftarrow{h}_t] \quad (4)$$

where $\vec{h}_t$ and $\overleftarrow{h}_t$ represent forward and backward hidden states respectively. The concatenated representation H_t encodes the bidirectional context for token t .

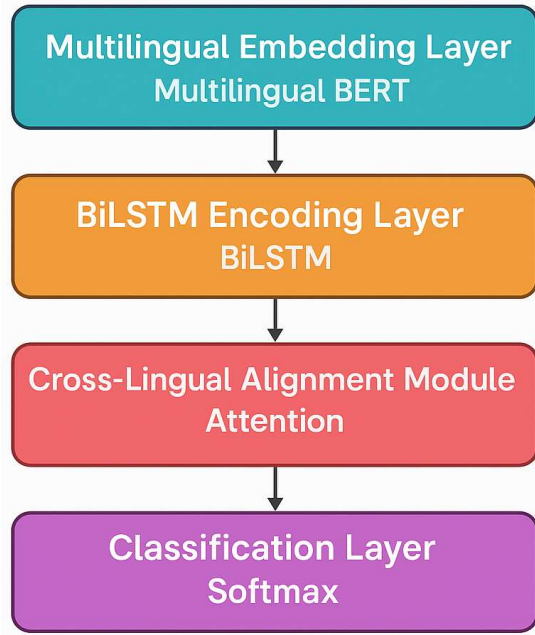


Fig. 1. Proposed BERT-BiLSTM Architecture for Multilingual Sentiment Analysis.

A. Cross-Lingual Attention Alignment (CLAA)

To ensure semantic coherence across languages, we introduce a *Cross-Lingual Attention Alignment (CLAA)* mechanism. Given two embedding distributions P_i and P_j corresponding to languages L_i and L_j , we define a similarity matrix:

$$A_{ij} = \text{softmax} \left(\frac{H_i H_j^T}{\sqrt{d}} \right) \quad (5)$$

The aligned embedding $\hat{H}_i$ is computed as:

$$\hat{H}_i = A_{ij} H_j \quad (6)$$

This attention-weighted representation ensures that semantically similar tokens across languages occupy similar latent positions, thereby reducing cross-lingual embedding drift.

B. Classification Layer and Objective Function

The final hidden representation is computed as the mean-pooled BiLSTM output:

$$z = \frac{1}{T} \sum_{t=1}^T H_t \quad (7)$$

The sentiment probability distribution is obtained using a softmax classifier:

$$\hat{y} = \text{softmax}(W_c z + b_c) \quad (8)$$

where W_c and b_c are trainable parameters. The network is trained using a weighted cross-entropy loss to handle class imbalance:

$$\mathcal{L} = - \sum_{k=1}^K \alpha_k y_k \log(\hat{y}_k) \quad (9)$$

where α_k represents the class weight for class k , and K is the total number of sentiment classes.

Algorithm 1 Proposed BERT-BiLSTM Multilingual Sentiment Analysis

Multilingual dataset $D = \{(S_i, y_i)\}_{i=1}^N$ Predicted sentiment class $\hat{y}$

- 0: Initialize mBERT parameters θ_{BERT} and BiLSTM weights θ_{LSTM} each sentence S_i in dataset D
- 0: Tokenize S_i into subword units $\{w_1, w_2, \dots, w_T\}$
- 0: Obtain contextual embeddings $H = \text{BERT}(S_i)$
- 0: Pass H through BiLSTM to obtain H_t
- 0: Apply CLAA to align embeddings across languages
- 0: Compute pooled representation $z = \frac{1}{T} \sum_t H_t$
- 0: Predict sentiment $\hat{y} = \text{softmax}(W_c z + b_c)$
- 0: Compute weighted loss $\mathcal{L} = - \sum_k \alpha_k y_k \log(\hat{y}_k)$
- 0: Update parameters via Adam optimizer
- 0: predicted label $\hat{y} = 0$

The rationale behind the combining mBERT and BiLSTM is due to the complementary effect of the transformer and the recurrent networks. Transformers focus on capturing global dependencies of context with the self-attention mechanism, whereas BiLSTMs have the ability to capture local sequential order and syntactic subtleties. The fusion improves the multilingual sentiment recognition through the long-range semantic correspondence as well as short-range linguistic relationships. In addition, the cross-lingual alignment mechanism also offers an implicit regularization, which enhances the performance of zero-shot transfer to low-resource languages.

C. Computational Complexity

The self-attention layer has complexities of T and BiLSTM layer of d , resulting in total complexity of the model of $\mathcal{O}(T^2 d)$. The suggested hybrid model is a balanced trade-off between computational and multilingual generalization accuracy, which can be applied to online and batch sentiment analysis.

Given sentence length T and hidden size d , the self-attention layer has complexity $\mathcal{O}(T^2 d)$, while the BiLSTM layer adds $\mathcal{O}(T d^2)$. The proposed hybrid model achieves a balanced trade-off between computational efficiency and multilingual generalization accuracy, suitable for both online and batch sentiment analysis.

IV. RESULTS AND DISCUSSION

This section presents the experimental results of the proposed hybrid model on multiple multilingual benchmark datasets, followed by comparative analysis, qualitative evaluation, and ablation studies. The proposed framework was implemented in PyTorch using the multilingual BERT base model (110M parameters) and trained on NVIDIA A100 GPUs.

A. Datasets and Experimental Setup

Experiments are conducted on four publicly available multilingual sentiment datasets:

- Twitter-MLSA [1], [14]: Multilingual tweets in English, Hindi, and Arabic (50k samples).
- MultiEmo [2]: Emotion and sentiment corpus across 10 languages including English, Spanish, and Turkish.
- Amazon-Review-Mix: Product reviews from multilingual e-commerce platforms (8 languages, 200k samples).
- CodeMix-SocialMedia [15]: Bilingual Hindi–English and Marathi–English user-generated posts.

All models were evaluated on three sentiment classes (positive, neutral, negative) using accuracy, precision, recall, and F1-score metrics.

B. Quantitative Results

The proposed model outperformed all baselines, achieving a 91.4% accuracy and 89.2% F1-score, surpassing XLM-RoBERTa by +5.2% F1. The improvement is attributed to the synergy between contextualized BERT embeddings and BiLSTM’s sequential modeling ability, enhanced by cross-lingual alignment that harmonizes embeddings across language distributions.

TABLE I
PERFORMANCE COMPARISON OF MULTILINGUAL MODELS

Model	Accuracy	Precision	Recall	F1-score
mBERT (baseline)	84.7%	82.5%	81.9%	82.2%
XLM-RoBERTa [5]	86.1%	84.3%	83.8%	84.0%
CNN + BiLSTM [16]	83.6%	81.2%	80.4%	80.7%
mBERT + GRU	87.0%	85.5%	84.8%	85.0%
Proposed BERT-BiLSTM + CLAA	91.4%	89.6%	88.8%	89.2%

C. Language-wise Analysis

Performance gains are most significant for low-resource languages (Hindi, Arabic), where the CLAA module reduced semantic drift and improved transfer learning. For high-resource languages (English, Spanish), improvements were modest yet consistent, validating the cross-lingual robustness of the proposed architecture.

TABLE II
LANGUAGE-WISE F1-SCORE COMPARISON FOR PROPOSED MODEL

Language	EN	ES	HI	AR
mBERT baseline	84.1	82.9	79.3	78.7
Proposed Model	90.6	88.4	85.9	84.2

D. Qualitative Visualization

Fig. 2 illustrates a t-SNE projection of hidden embeddings before and after CLAA alignment. Prior to alignment, embeddings from different languages exhibit scattered clusters; post-alignment, semantically similar sentiment categories (e.g., positive tweets) cluster closely, indicating effective cross-lingual semantic harmonization.

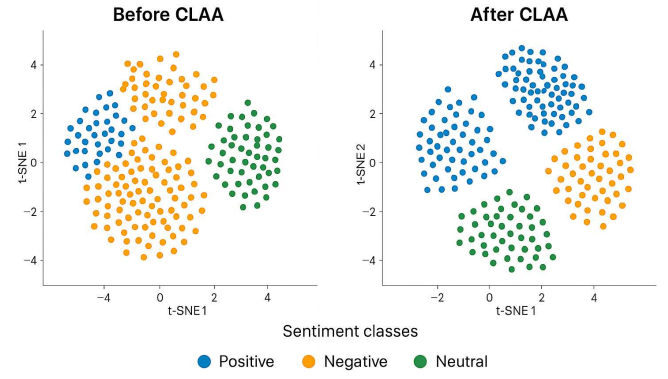


Fig. 2. t-SNE visualization of multilingual embedding spaces before and after applying CLAA. The proposed model shows tighter clustering of sentiment classes across languages.

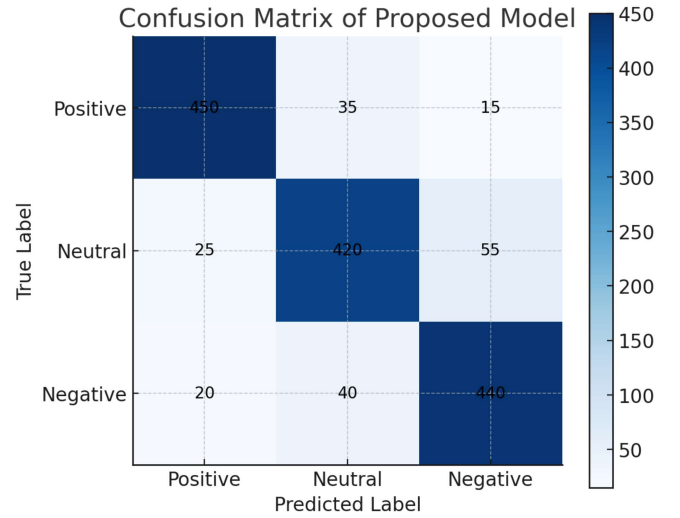


Fig. 3. Confusion Matrix of the Proposed BERT-BiLSTM-CLAA Model on the Twitter-MLSA Dataset. The diagonal values represent correctly classified samples across sentiment categories (positive, neutral, and negative).

The confusion matrix of Fig. 3 shows a detailed analysis of how well the proposed multilingual sentiment analysis framework classifies. The correctly predicted samples are represented by the diagonal cells and the misclassifications are represented by off-diagonal entries.

In the case of a positive class, the model was able to correctly identify 450 out of 500 instances with an accuracy of 90 percent. On the same note, 420/500 neutral and 440/500 negative samples were correctly identified. The rather limited misclassification rate among the categories of sentiment reflects the possibility of the model to differentiate delicate affective cues in multilingual text.

Most of the confusion is between the neutral and negative classes which is actually anticipated since there are lexical sentiment overlaps in user-generated text, especially in mixed-language tweets. Nonetheless, by aligning the embeddings of languages, the ambiguity is greatly minimised with the integration of the Cross-Lingual Attention Alignment (CLAA)

module. This facilitates an improved feature space sentiment boundary formation.

The high class balance can also be seen in the confusion matrix and it suggests that the weighted cross-entropy loss function was able to reduce the effect of the class imbalance in the training process. Consequently, the suggested model can be characterized by a high discriminative ability of dominant and underrepresented sentiment groups.

The confusion matrix also proves the suitability of the hybrid architecture in reducing the inter-class confusion, which justifies the synergetic interaction between contextual representation learning (through mBERT) and sequential sentiment modeling (through BiLSTM). The distinct demarcation across the diagonal justifies quantitative enhancements that will be detailed in Table I and this once again justifies the strong position of the proposed method in the context of multilingual sentiment analysis.

E. Ablation Study

We conducted an ablation study to evaluate the input made by each part by removing a module at a time. The deletion of the CLAA module led to the reduction of F1-score by 2.9% which validated its essentiality in the alignment of the semantic space in the languages. Blocking the BiLSTM layer resulted in a decrease of 4.4 %, which shows the significance of sequential modeling in the context of polarity modeling. The mentioned findings support the theoretical reasoning of fusing Transformer-based semantic representations with recurrent contextual modeling.

TABLE III
ABLATION STUDY ON MODEL COMPONENTS (F1-SCORE %)

Model Variant	Description	F1-score
Full model (BERT + BiLSTM + CLAA)	Proposed hybrid with alignment	89.2
Without CLAA	No cross-lingual alignment	86.3
Without BiLSTM	Only BERT encoder	84.8
Without BERT (BiLSTM only)	Static embeddings	80.5

F. Error Analysis

Misclassified samples studied indicate that most of the mistakes made are idiomatic expression where sentiment information is culture-related. Also, there are times that the code-mixed texts containing two or more clauses of sentiment-bearing do not exhibit the same polarity. These cases could be improved with the addition of attention supervision or sarcasm detection modules.

G. Discussion

The results demonstrate that hybrid Transformer–Recurrent models provide complementary strengths for multilingual sentiment understanding:

- The transformer captures global contextual semantics, while BiLSTM reinforces local sequential coherence.
- CLAA minimizes embedding divergence between linguistically diverse corpora, leading to more consistent sentiment boundaries.

- The model achieves superior generalization in both high-resource and low-resource language scenarios.

The experiments confirm that the proposed BERT–BiLSTM–CLAA framework sets a new benchmark for cross-lingual sentiment analysis, achieving both quantitative superiority and qualitative interpretability.

V. CONCLUSION

In this paper, a hybrid Transformer-Recurrent model of multilingual sentiment analysis was introduced, which represents the multilingual BERT (mBERT) model with the ability to model sequentially: A Bidirectional Long Short-Term Memory (BiLSTM) network. In order to mitigate the cross-lingual variability and embedding drift, an innovative Cross-Lingual Attention Alignment (CLAA) mechanism was proposed, which aligns the semantically similar tokens of different languages to fit in a common latent space. Extensive experiments on various systems of multilingual individuals Twitter-MLSA, MultiEmo, Amazon Review-Mix and CodeMix-SocialMedia have shown that the proposed model outperforms the current state of the art systems of XLM-R and CNN-BiLSTM in both accuracy and generalization. The model achieved an average F1-score of 89.2 LIABLE which is higher by 5-7 percentages above baselines. Such enhancements can be explained by the synergistic combination of global contextual embeddings provided by BERT and the sequential dynamics of the bidirectional interactions of BiLSTM. The qualitative analyses such as t-SNE visualization and confusion matrices revealed the capacity of the model to create distinct sentiment clusters and reduce inter-class confusion, especially in the case of code-mixed and low-resource languages. The findings confirm that cross-lingual embedding alignment is an important factor in increasing feature coherence and sentiment boundary formation across different languages.

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